

An Algorithm For Detection of Nutritional Deficiencies from Digital Images of Coffee Leaves Based on Descriptors and Neural Networks

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Abstract— This work proposes an algorithm for detection of nutritional deficiencies of coffee plantations from the analysis of tonalities and geometric characteristics of the leaves. The algorithm aims to reduce the subjectivity in the analysis by visual perception. Errors in this analysis affect the dosage plan of fertilizers and nutrients applied by producers. The algorithm is formed first by a step of contrast improvement from the luminance followed by a SIFT algorithm that provides the important points for the generation of the corresponding descriptors. In parallel to this, the improved image is subjected to thresholding to obtain Hu and Fourier descriptors. With the three types of descriptors, a specific neuronal network is trained separately according to the nutritional deficiency to be detected. Kappa index was used to compare the results with those taken by visual inspection. The results were satisfactory, obtaining a Kappa coefficient of 0.96 for N and P deficiency, and 0.92 for B.

Keywords– Neural Networks, Image Processing Algorithm SIFT descriptors, Hu, Fourier, Coffee, Leaves

I. INTRODUCTION

In recent years, Precision Agriculture (PA) has begun to take off, thanks to the acceptance of farmers and organizations that help agricultural research. This type of agriculture is based on the collection of information about the crops so that they can take better measures in the face of problems such as lack of fertilizer or water, diseases or pests. To be able to carry out a sustainable agriculture, it is necessary that the PA be used, and that the technology improves the agricultural techniques. This can be achieved through the analysis of the information collected by sensors, images, GPS and other gadgets [1].

Particularly in the cultivation of coffee, one of the biggest problems for producers is the precise classification of nutritional deficiencies in coffee plants, since there is no automatic method defined by which you can find the type of malnutrition suffered by a plant. For the most part, diagnostics are done by simple visual tests, which are usually inaccurate, since they are based on the experience of each agronomist expert, which differs greatly according to the plantation and its expertise. For a more accurate diagnosis, a considerable amount of leaves should be sent, as a sample of the coffee plant, to a laboratory so it can be tested. The leaves should be put on ice and send to the laboratory to perform a foliar chemical analysis within 2 days that it was separated from the coffee plant, so it does not wither. The results

of the tests are sent by mail in an average time of 1 week. This is a more accurate diagnosis since it is based on the chemistry of the leaf, however it is a process that takes a long time and has a high cost, which makes the analysis of all the plants of the crop difficult [2].

Currently there are systems models that work with image processing for the detection or recognition of diseases on fruits or plants, by using Scale Invariant Feature Transform (SIFT) [3] to find important points in the images, which are then classified depending on what part of the image they find and then they can use to make comparisons with other images and find similarities [3]. However, this type of processing in addition to taking a lot of time is not always the right one for places where the lighting is not the same throughout the day.

Other solutions that have been tried are the use of drones for the collection of crop information through a video camera that at the same time is processing the images to obtain information that can be used by farmers to improve their crops. In addition to the ease of collection through drones, the sensors attached to these are connected to other IoT (Internet of Things) devices, which help with processing and, later, with the tracking of improvements made by the farmer [4]. Despite this, this type of project that can work where the terrain is mostly flat, the strength of the winds plays an important factor in that the drone could not be handled in the same way. In addition to the remote places where most of the crops are located, they do not have access to the internet easily, which would make it difficult to collect data in an efficient way.

This paper its composed of several elements for the proposed work: for detecting the deficiencies like nitrogen and phosphorus, which depends on colors, SIFT is used to find the key points in the image based on the colors; for the detection of the calcium deficiency, which depends on the shape of the edge of the leaf, Fourier descriptors were used; and finally, for the boron deficiency, Hu's moments were used to describe the shape of the leaf. This information was processed by an already trained neural networks with a database extracted from the same production area and the algorithm has been validated with the expert evaluation of two agronomists.

In reminder, Section 2 details the proposed method mechanism; Section 3 presents this method results and Section 4 contains concluding remarks.

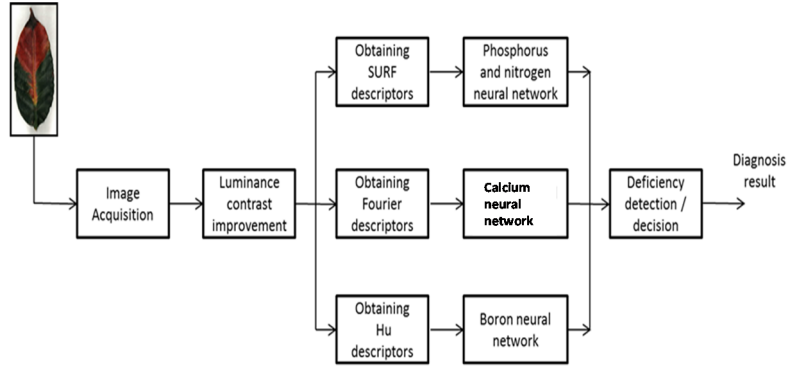


Fig. 1 Block diagram of the proposed algorithm with all the descriptors and neural networks.

II. DESCRIPTION OF THE PROPOSED METHOD

The block diagram shows the process that the image of the leaf is going to pass to obtain its features through different methods to predict its deficiency as is shown in Fig. 1, which will be detailed in the following sections.

A. Nutritional deficiencies in coffee leaves

The following describes the visible symptoms of four nutritional deficiencies:

Nitrogen: They appear first in the oldest leaves (in the lower branches of the coffee tree). A uniform chlorosis is seen advancing from the base towards the apex of the leaf and from the central vein to the edges of the leaf. When the deficiency is more severe, it becomes more chlorotic and covers the entire limbus.

Phosphorus: Lime green lobar chlorosis is exhibited between the veins of the oldest leaves. These leaves have irregular yellow-tan spots, as well as areas with a reddish coloration.

Calcium: They are present in the younger leaves (in the upper part of the plant). These present deformation and ripple of the edges.

Boron: They manifest in the younger leaves of the plant. These have small size, irregular and misshapen edges. In turn, they look long, crooked and / or wrinkled. They present partial or total loss of the apex.

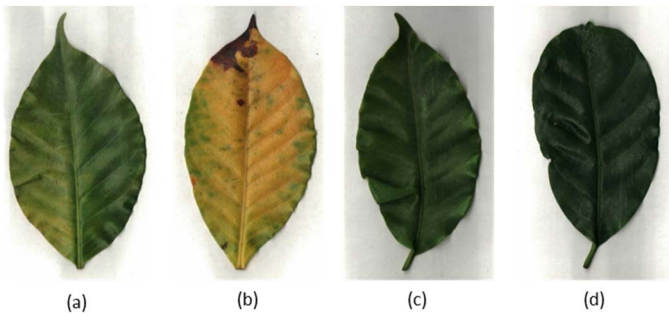


Fig. 2 Nutritional deficiencies: (a) Nitrogen, (b) Phosphorus, (c) Calcium and (d) Boron.

B. Acquisition of the image

For image acquisition, a portable scanner (VuPoint Magic Wall; model PDSWF-ST47-VP) was chosen.

It comes with an internal battery with an approximate duration of 5 h (700 mA h). It is possible to take this system to the field.

The scanner possesses three grades in resolution (1200, 600, and 300 dpi). The 300-dpi resolution was used, in color, (i) because this resolution is sufficient to obtain the pertinent characteristics of the sheet for processing and (ii) because the scanning time decreases because of the lower resolution (to 3 s). This resolution, for the maximum size of the coffee leaf (105×148 mm), is 2100×1500 pixels. The scanner obtains images in the JPEG format and collaborates with a contact image sensor that optimizes the resolution and energy expended. The dimensions of the scanner are $31.5 \times 14 \times 5.1$ cm with a weight of 480g. In addition, the scanner used a micro secure digital (SD) card, which permitted the transfer of images to the Raspberry Pi [5]. The scanner also possesses a liquid crystal display (LCD) screen, so that the user is able to monitor whether the image was scanned correctly, maintain the sheet count, and verify the battery supply.

C. Enhancing luminance contrast

At the preprocessing stage, imperfections in the image, such as those caused by dust or external agents, were removed to avoid confounding the image processing. First, the image that was captured in the RGB mode with primary components $I_R(x, y)$, $I_G(x, y)$, and $I_B(x, y)$ was subjected to thresholding to identify objects that required removal. The threshold is expressed as:

$$I'_R(x, y) = \begin{cases} I_R(x, y) & ; \quad U_0 < I_R(x, y) \leq U_1 \\ 255 & ; \quad \text{in other cases} \end{cases} \quad (1)$$

$$I'_G(x, y) = \begin{cases} I_G(x, y) & ; \quad U_0 < I_G(x, y) \leq U_1 \\ 255 & ; \quad \text{in other cases.} \end{cases} \quad (2)$$

$$I'_B(x, y) = \begin{cases} I_B(x, y) & ; \quad U_0 < I_B(x, y) \leq U_1 \\ 255 & ; \quad \text{in other cases.} \end{cases} \quad (3)$$

where U_1 is the maximum threshold value and U_0 is the value of the minimum threshold. In this case $U_1 = 127$, and which were obtained $U_0 = 10$, to minimize the error of segmentation of the regions of interest, as seen in Fig. 3, these were chosen according to the tests carried out with the scanner. Then, starting from $I'_R(x, y)$, $I'_G(x, y)$ and $I'_B(x, y)$ you get the luminance component $I_L(x, y)$ that was obtained with the CIE (Commission Internationale de l'Éclairage) color model $L^*a^*b^*$ [6]. This component was then subjected to a contrast enhancement to highlight the

regions of interest. The contrast limited adaptive histogram equalization technique was applied [7], where the histogram made a cut at a predefined value and then applied the general formula for equalization.



Fig. 3 A leaf before and after noise elimination.

Thus, the equalization correspondence function can be expressed as:

$$h(v) = \text{round} \left(\frac{\text{pdf}(v) - \text{pdf}_{\min}}{(M \times N) - \text{pdf}_{\min}} \times (L_g - 1) \right) \quad (4)$$

where v is the input luminance value, $\text{pdf}(v)$ is the probability distribution function of $I_L(x, y)$, M is the number of rows in the image, N is the number of columns and L_g is the number of gray levels used. The preset minimum value of the distribution function was $\text{pdf}_{\min} = 3$, according to the tests carried out. Then, $I'_L(x, y)$ was defined as the resulting image of equalization. The RGB model was then returned from the equalized luminance and the color components a^* and b^* . The primary components obtained were referred to as $I_{2R}(x, y)$, $I_{2G}(x, y)$, and $I_{2B}(x, y)$.

D. Obtaining descriptors SIFT

To identify the strongest points of interest in the RGB image obtained, the image was converted into grayscale $I_g(x, y)$. Then, $I_g(x, y)$ was applied to the SIFT [8]. The inputs were the image $I_g(x, y)$ and the maximum number of points of interest, which was established as 500, to lessen the computational burden. From the algorithm, K points of interest were obtained, and the value of response $R_e(x, y)$ of each point of interest and the descriptors of each point were expressed as $d_i(x, y)$, for $i = 0, 1, \dots, 127$. Thus, 128 descriptors were determined for each point of interest. Next, the points were sorted from highest to lowest according to the value of $R_e(x, y)$. In that order, the neighboring points of interest and their descriptors in the spatial regions within 5×5 pixels were eliminated. Through this method, only the greatest points of interest were maintained, without repetition.

E. Obtaining Fourier descriptors

To utilize the Fourier descriptors, the gray image $I_g(x, y)$ underwent a threshold filter to determine the shape of the leaf inside the image. The thresholding is expressed as:

$$I_b(x, y) = \begin{cases} 1; & I_g(x, y) > U_2 \\ 0; & I_g(x, y) < U_2 \end{cases} \quad (5)$$

where U_2 is the threshold and $I_b(x, y)$ is the image obtained. The threshold value of this process was $U_2 = 200$. This value

was chosen according to the tests carried out with the scanner, as indicated in Fig. 4.



Fig. 4 The image of leaf before (left) and after (right) thresholding of $U_2 = 200$.

Next, the outline of the largest object in $I_b(x, y)$, which would be (the leaf) was obtained. This $I_b(x, y)$ was utilized in the algorithm proposed by Suzuki [9], as the algorithm provides the fastest and most accurate results. The output of the algorithm is the coordinates (x, y) of the contours detected. However, only the coordinates of the leaf were retained. To obtain the Fourier descriptors from the leaf, a fast Fourier transform [10] was utilized at size 200. The input values to this algorithm were the coordinates of the pixels of the outline, expressed in the complex form $x + jy$. The magnitude values of each Fourier coefficient obtained, $F(u)$ (for $u = 0, 1, \dots, 199$), were a part of the descriptors that were utilized as inputs to the neural networks.

F. Obtaining Hu descriptors

To obtain the Hu descriptors, the outline of the leaf was detected by obtaining a binary image and its corresponding coordinates (x, y) . With these values, the moments of Hu were expressed as mh_n for $n = 1, 2, \dots, 7$ [11]. Finally, to obtain better results in the comparison of characteristics of the objects, the following logarithmic transformation was applied to the moments obtained:

$$mh'_n = \frac{mh_n}{\sqrt{mh_n \times mh_n}} \times \log_{10}(|mh_n|) \quad (6)$$

G. Phosphorus and Nitrogen neural network

To generate the artificial neural networks for phosphorus [12] and nitrogen, the descriptors were inputted into the SIFT. Studies have concluded that both networks must be composed of six layers (one input, four hidden and one output layer). The inputs are the number of descriptors for each image; hidden layer 1 is composed of 500 neurons; hidden layer 2, 200 neurons; hidden layer 3, 100 neurons; hidden layer 4, 50 neurons; and for the output layer, two outputs are utilized. For the neuron transfer function, there were five options: (1) sigmoid, (2) tangent arc, (3) hyperbolic tangent, (4) normalized tangent arc, and (5) normalized hyperbolic tangent. The order of the nitrogen transfer of the neuronal network was (consecutively): 3, 3, 1, 1, and 1, for the five layers. The best approximation results were obtained through this sequence.

The appropriate λ (adaptive value that determines the trend of the algorithm) was chosen so that generalization of the data was not reduced. Therefore, $\lambda = 0.4$. This helped, ensuring that the neural network produced accurate results. The network was utilized after 1000 iterations of training, after which there is no significant improvement. Finally, in the output neurons, the higher percentage was chosen from the two values. Depending on the position, the values were "0" (without deficiency) or "1" (with deficiency).

H. The neural network of calcium

The same procedure was followed as that of the creation of the neural network for nitrogen. The entries were the Fourier descriptors of the outline of the leaf. The following structure was utilized: hidden layer 1 was composed of 400 neurons; hidden layer 2, 220 neurons; hidden layer 3, 100 neurons; and hidden layer 4, 80 neurons. There were two outputs to be considered. The transfer function possessed the same options as for nitrogen: 3, 3, 1, 1, and 1. The most accurate approximations were obtained. The value was $\lambda = 0.17$. The number of iterations was also the same as that of the nitrogen neural network.

I. The neuronal network for Boron

The procedure was similar to that of the other elements. The descriptors of Hu were the input layer. For hidden layer 1, 450 neurons were utilized; for hidden layer 2, 300 neurons; for hidden layer 3, 200 neurons; and for hidden layer 4, 50 neurons. Two outputs were considered as the last layer. For the transfer function, the following sequence was utilized: 3, 3, 1, 1, and 1. We found that $\lambda = 0.25$ and the number of iterations was, again, the same as previous networks.

J. Deficiency Detection/Decision

Each neural network provided a value of “1” or “0”. If the value was “1”, the processed image indicated a deficiency from its corresponding network. If the result was “0”, there was no deficiency in that network. Each image produced results for each deficiency in the following order: boron, calcium, phosphorus, and nitrogen.

K. Procedure for the taking of samples

The type of coffee evaluated was Catimor, an Arabic coffee variety. The samples were taken from Cenfrocafe plantation, in Jaen, Cajamarca - Peru. A total of 75 images were collected, and deficiency was detected in 70 leaves. The following procedure was executed to validate the networks:

Step 1: A specialist visually inspected the leaves and recorded their deficiencies.

Step 2: The same procedure was repeated by a second specialist. Both specialists were expert agronomists specialized in nutritional deficiencies of coffee leaves.

Step 3: The leaves with identical evaluations between the experts were used as references.

Step 4: The deficiency of the leaf was evaluated with the algorithm.

Step 5: The presence or lack of concordances between the specialists and the software were recorded.

Concordance measurements (per deficiency) were performed through Cohen's Kappa index [13].

III. RESULTS

In order to train the neural networks, images from the scanning procedure were utilized. The results are shown in Table I. Note that the closer the value is to 1, the higher the level of concordance between the specialists and the algorithm.

Therefore, the results are satisfactory for the cases of boron, nitrogen, and phosphorus, but there is a lack of concordance with

calcium. This is because the amount of leaves that did not have a calcium deficiency was low compared to the number of leaves that possessed a calcium deficiency. Therefore, there were not enough references for sufficient training of the network.

For validation, there was the support of two agronomists' experts in nutrient deficiencies in coffee leaves. Each specialist evaluated the sample leaf by a visual inspection, without knowing the outcome of algorithm proposed or the result of his colleague.

TABLE I. RESULTS FROM COHEN KAPPA INDEX VALIDATION

Nutritional deficiency	Index Kappa Cohen
Nitrogen	0.968351441
Phosphorus	0.962674095
Calcium	0.316326531
Boron	0.924972004

Table I shows the coefficients obtained with the algorithm are satisfactory. For Nitrogen, Phosphorus and Boron the coefficient is above 0.9 which means that there is qualitatively an almost perfect concordance between the results of the algorithm and the experts. In the case of Calcium, it is different, since it was lower than 0.5, which gives it an acceptable rating, this is because the samples taken were from coffee crops in the North, where the land is acidic and therefore has a deficit of Calcium, the algorithm does not recognize Calcium leaves well since almost all the leaves have Calcium deficiency, but the experts do not take it into account since it is something normal in the crops of that region.

IV. CONCLUSIONS AND FUTURE IMPROVEMENTS

An algorithm was developed with the purpose of detecting nitrogen, phosphorus, boron and calcium deficiencies in coffee leaves, in order to provide an in-situ result for these nutrient deficiencies. The image processing techniques, from image acquisition to diagnosis result, offer an accuracy of 96% for detection of nitrogen; 96% for phosphorus; and 94% for boron deficiencies. Thus, the algorithm presents excellent performance with respect to existing methods.

A higher accuracy for calcium deficiency detection could be achieved by collecting samples from different plantations not only in Cajamarca, but in other areas in Peru that has coffee plantations. This would increase the number of healthy coffee leaves samples, therefore it would be more notorious the characteristics of calcium deficiencies in leaves that in other cases would be considered healthy.

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